

Analysis

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TPOT with linear search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##   task_id .y. group1      group2  n1  n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int>    <dbl>    <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~ 21  21    198 2.86e-3 **
## 2  146818 auroc ensemble_random_~ indiv~ 21  21    189 9.02e-3 **
## 3  146820 auroc ensemble_random_~ indiv~ 21  21    187 1.13e-2 *
## 4  168350 auroc ensemble_random_~ indiv~ 21  21    225 1.34e-5 ****
## 5  168757 auroc ensemble_random_~ indiv~ 21  21    222 3.15e-5 ****
## 6  168784 auroc ensemble_random_~ indiv~ 20  20    207 9.54e-6 ****
## 7  168911 auroc ensemble_random_~ indiv~ 21  21    194 4.88e-3 **
## 8  190137 auroc ensemble_random_~ indiv~ 21  21    206 8.52e-4 ***
## 9  190146 auroc ensemble_random_~ indiv~ 20  20    138 2.31e-1 ns
## 10 190411 auroc ensemble_random_~ indiv~ 21  21    228 4.77e-6 ****
## 11 359955 auroc ensemble_random_~ indiv~ 21  21    157 1.54e-1 ns
## 12 359956 auroc ensemble_random_~ indiv~ 21  21    189 9.02e-3 **
## 13 359958 auroc ensemble_random_~ indiv~ 19  19    169 3.1 e-3 **
## 14 359959 auroc ensemble_random_~ indiv~ 21  21    216 1.31e-4 ***
## 15 359960 auroc ensemble_random_~ indiv~ 21  21     11 1 e+0 ns
## 16 359962 auroc ensemble_random_~ indiv~ 21  21    209 5.1 e-4 ***
## 17 359963 auroc ensemble_random_~ indiv~ 21  21    218 8.39e-5 ****
## 18 359964 auroc ensemble_random_~ indiv~ 21  21    210 4.26e-4 ***
## 19 359965 auroc ensemble_random_~ indiv~ 21  21    184 1.58e-2 *
## 20 359968 auroc ensemble_random_~ indiv~ 21  21    181 2.16e-2 *
## 21 359969 auroc ensemble_random_~ indiv~ 21  21    231 9.54e-7 ****
## 22 359970 auroc ensemble_random_~ indiv~ 21  21    230 1.91e-6 ****
## 23 359971 auroc ensemble_random_~ indiv~ 21  21    212 2.93e-4 ***
## 24 359972 auroc ensemble_random_~ indiv~ 21  21    229 2.86e-6 ****
## 25 359974 auroc ensemble_random_~ indiv~ 21  21    197 3.28e-3 **
## 26 359975 auroc ensemble_random_~ indiv~ 21  21    205 1.98e-3 **
```

Medians

```
## # A tibble: 26 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.875          0.890
## 2  146818          0.935          0.935
## 3  146820          0.997          0.997
## 4  168350          0.963          0.966
## 5  168757          0.785          0.795
## 6  168784          0.966          0.968
## 7  168911          0.873          0.874
## 8  190137          0.918          0.930
## 9  190146          0.962          0.964
## 10 190411          0.920          0.922
## 11 359955          0.742          0.748
## 12 359956          0.930          0.934
## 13 359958          0.939          0.942
## 14 359959          0.743          0.754
## 15 359960          1          1
## 16 359962          0.836          0.846
## 17 359963          0.995          0.996
## 18 359964          0.993          0.994
## 19 359965          1.00          1.00
```

##	20	359968	0.919	0.929
##	21	359969	0.847	0.851
##	22	359970	0.910	0.921
##	23	359971	0.997	0.997
##	24	359972	0.990	0.993
##	25	359974	0.830	0.873
##	26	359975	0.993	0.996